
Universal Plastic Shredder V2.0

Weekly Progress Reports

PORTLAND STATE UNIVERSITY
MASEEH COLLEGE OF ENGINEERING & COMPUTER SCIENCE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Course: ECE 412 / 413 Senior Capstone
Term: Winter–Spring 2026
Sponsor: MME / Ichor Systems (Bridget Lannigan)
EE Advisor: Dr. Jonathan Bird
Team: Bao Nguyen, Fearghus Tyler,
Yaqoub Rabiah, Fox Kang

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1 Universal Plastic Shredder V2.0 — Weekly Progress Reports

Course: ECE 412/413 (Spring 2026) . **Sponsor:** MME / Ichor Systems **Team:** Bao Nguyen, Fearghus Tyler, Yaqoub Rabiah, Fox Kang

1.1 Key hardware referenced

System	Part	Role / interface
VFD	Huanyang HY02D211B-T	Motor drive. Speed ref on VI (0–10 V, PD002=1), current monitor on VO (PD054=1), fault relay FA/FC (PD052=02), run inputs FOR/REV/RST (sinking to DCM).
Motor	GE 5KE182BC205B (bench) / PE145TC-2-4 (production cand.)	3-phase, FLA ~ 7.6 A. Phases U/V/W -> T1/T2/T3 via 4 mm banana sockets.
PLC	DirectLOGIC 205, H2-DM1E CPU	DI X0–X7 (D2-08ND3), relay out Y0–Y7 (D2-08TR), current-in F2-08AD-1 , 0–10 V out F2-08DA-2 .
HMI	C-More EA1-T4CL	4” touch panel, serial link to CPU.
Contactor	Mitsubishi SD-N35	In the hardwired E-stop chain (coil A1/A2); switches the VFD feed.
E-stop	Eaton E22B1 (NC)	Drops the contactor coil; state monitored on PLC X4 .
Braking resistor	ATO APCS-300R30-AD	300 W / 30 ohm on VFD P+/PR .

1.2 Report index

Winter term

Date	Focus
2025-12-17	Kickoff coordination meeting; component research begins
2026-01-03	Compare motors / HMIs / safety sensors
2026-01-09	Sponsor + advisor goal/budget review
2026-01-16	Motor-control & safety strategy; first motor+VFD bench test
2026-01-23	Electrical design starts; BOM for initial purchases; AutoCAD setup
2026-02-02	Prototype build; safety-sensor circuit
2026-02-06	Schematic + ME coordination
2026-02-13	Advisor coordination meeting
2026-02-27	Documentation & presentation prep; BOM revision
2026-03-06	Parts arrived — fit & wire the prototype
2026-03-13	VFD/motor bring-up; HMI GUI
2026-03-16	Finalize documentation before spring break

Spring term

Wk	Date	Focus
1	2026-04-03	Post-break reset; points list; E-stop -> contactor
2	2026-04-10	PLC <-> circuit integration; cap-touch (small scale)
3	2026-04-17	Relay-module fix; analog modules; HMI scaling
4	2026-04-24	Current -> voltage (0–10 V) control swap; VFD reprogram
5	2026-05-01	Presentation; cap-touch prototype; braking resistor sizing
6	2026-05-08	Midterm week (light); documentation
7	2026-05-15	Enclosure panel build; overcurrent-signal logic
8	2026-05-22	Panel wiring; ME parts hand-off
9	2026-05-30	Documentation push (report, manual, schematics)

Wk	Date	Focus
10	2026-06-06	Final report & capstone demo

TERM
Winter Term
Dec 2025 – Mar 2026

Universal Plastic Shredder V2.0 – Weekly Progress Reports

WEEKLY REPORT 2025-12-17

Last week . (*hours in parentheses*)

Team Review - The electrical and mechanical teams held the first coordination meeting for the shredder project. - We discussed the overall system design, safety systems, and motor-control ideas. - The team also set up communication tools and began researching components.

Bao Nguyen - Created Discord server for team communication. (1) - Attended coordination meeting with ME team. (1) - Began researching HMI display options. (3)

Fearghus Tyler - Discussed motor requirements with mechanical team. (2) - Attended coordination meeting with ME team. (1) - Researched possible motor options. (2)

Yaqoub Rabiah - Started researching VFD motor-control systems. (3) - Attended coordination meeting with ME team. (1)

Fox Kang - Started researching safety sensors and safety concepts. (3) - Attended coordination meeting with ME team. (1)

Next week

Team Plan - Continue researching components and begin building comparison tables for motors, sensors, and control systems.

Bao Nguyen - Research HMI options and telemetry display ideas. (3)

Fearghus Tyler - Begin creating preliminary electrical BOM. (3)

Yaqoub Rabiah - Research VFD telemetry and monitoring features. (3)

Fox Kang - Continue researching safety detection methods. (3)

Blocked

Bao Nguyen - Waiting for motor requirements from mechanical team.

Fearghus Tyler - Waiting for gearbox specifications.

Yaqoub Rabiah - Waiting for VFD model.

Fox Kang - None.

WEEKLY REPORT 2026-01-03

Last week . (*hours in parentheses*)

Team Review - The team continued researching electrical components for the shredder system. - Motors, HMIs, and safety sensors were compared. - Early ideas for the electrical control system were discussed.

Bao Nguyen - Researched HMI display options. (3) - Reviewed possible telemetry information to display. (2)

Fearghus Tyler - Began building the electrical BOM for the initial test setup — motor, VFD, contactor, PLC rack + modules, HMI, breaker, power switch, and 12/10 AWG wire. (3)

Yaqoub Rabiah - Compared VFD options and specifications. (3)

Fox Kang - Continued safety-sensor research. (3)

Next week

Team Plan - Narrow down hardware options and begin planning electrical architecture.

Bao Nguyen - Work on system block diagram. (3)

Fearghus Tyler - Expand electrical BOM. (3)

Yaqoub Rabiah - Research PLC integration with VFD. (3)

Fox Kang - Continue researching safety interlocks. (3)

Blocked

Bao Nguyen - Waiting for enclosure dimensions.

Fearghus Tyler - Waiting for torque specifications.

Yaqoub Rabiah - Waiting for PLC to arrive.

Fox Kang - Stuck on figuring out the type of technology for the safety sensor.

WEEKLY REPORT 2026-01-09

Last week . (*hours in parentheses*)

Team Review - The team met with the faculty advisor and industry sponsor to review project goals.
- Budget, motor power, and safety requirements were discussed.

Bao Nguyen - Attended sponsor meeting. (1) - Researched HMI communication options. (3)

Fearghus Tyler - Continued the electrical BOM — added the Huanyang HY02D211B-T VFD, contactor, push buttons, PLC rack + modules, C-More EA1-T4CL HMI, circuit breaker, power switch, and 12/10 AWG wire. (3) - Attended sponsor meeting. (1)

Yaqoub Rabiah - Researched VFD control and monitoring options. (4)

Fox Kang - Continued researching safety sensors. (3)

Next week

Team Plan - Start developing electrical system architecture.

Bao Nguyen - Work on system layout diagram. (3)

Fearghus Tyler - Continue updating BOM. (3)

Yaqoub Rabiah - Compare VFD options. (3)

Fox Kang - Research safety-sensor placement. (3)

Blocked

Bao Nguyen - Waiting for motor specs.

Fearghus Tyler - Waiting for gearbox decision.

Yaqoub Rabiah - Waiting on PLC.

Fox Kang - Waiting on HMI.

WEEKLY REPORT 2026-01-16

Last week . (*hours in parentheses*)

Team Review - The electrical team reviewed motor-control strategies and safety systems. - Design ideas for the electrical system were discussed.

Bao Nguyen - Tested small motor and VFD setup. (4)

Fearghus Tyler - Worked with mechanical team on gearbox ideas. (3)

Yaqoub Rabiah - Reviewed VFD specifications. (3)

Fox Kang - Created accident-scenario safety flow chart. (3)

Next week

Team Plan - Continue developing electrical control-system design.

Bao Nguyen - Work on electrical control-circuit layout. (3)

Fearghus Tyler - Update electrical BOM. (3)

Yaqoub Rabiah - Continue VFD research. (3)

Fox Kang - Continue safety-system research. (3)

Blocked

Bao Nguyen - Waiting for gearbox decision.

Fearghus Tyler - Waiting for enclosure size.

Yaqoub Rabiah - Waiting on equipment to program PLC.

Fox Kang - Choosing the correct parts for the safety sensors.

WEEKLY REPORT 2026-01-23

Last week . (*hours in parentheses*)

Team Review - The team began developing the electrical design and continued coordination with the mechanical team.

Bao Nguyen - Worked on electrical system design. (3)

Fearghus Tyler - Updated the electrical BOM for the first round of purchases (VFD, contactor, PLC rack/modules, HMI). (3) - Set up the AutoCAD schematic drawings per these specs — learning the workflow: creating reusable component blocks, transferring blocks between drawings, and setting up the sheet / title-block templates.

Yaqoub Rabiah - Began PLC control-logic planning. (3)

Fox Kang - Continued safety-system research. (3)

Next week

Team Plan - Start building prototype circuits and system diagrams.

Bao Nguyen - Design electrical circuit diagram. (3)

Fearghus Tyler - Continue component sourcing. (3)

Yaqoub Rabiah - Develop PLC logic structure. (3)

Fox Kang - Develop safety-system concept. (3)

Blocked

Bao Nguyen - Waiting for motor specs.

Fearghus Tyler - None.

Yaqoub Rabiah - None.

Fox Kang - None.

WEEKLY REPORT 2026-02-02

Last week . (*hours in parentheses*)

Team Review - The team worked on prototype development and safety-system concepts.

Bao Nguyen - Began building electrical test rig. (4)

Fearghus Tyler - Worked on the electrical schematic in AutoCAD — built out component blocks and the power-distribution sheet. (3)

Yaqoub Rabiah - Developed PLC control logic. (3)

Fox Kang - Requested capacitive-sensor samples. (2)

Next week

Team Plan - Continue prototype testing and system development.

Bao Nguyen - Continue test-rig development. (3)

Fearghus Tyler - Improve schematic design. (3)

Yaqoub Rabiah - Continue PLC programming. (3)

Fox Kang - Develop safety-sensor circuit. (3)

Blocked

Bao Nguyen - Waiting for some parts to arrive.

Fearghus Tyler - None.

Yaqoub Rabiah - None.

Fox Kang - Waiting for sensor samples.

WEEKLY REPORT 2026-02-06

Last week . (*hours in parentheses*)

Team Review - Continued electrical design work and coordination with the mechanical team.

Bao Nguyen - Worked on wiring prototype components. (3)

Fearghus Tyler - Updated the AutoCAD schematic diagrams (block edits and layer cleanup). (3)

Yaqoub Rabiah - Continued PLC logic work. (3)

Fox Kang - Worked on safety-system design. (3)

Next week

Team Plan - Continue system testing and design improvements.

Bao Nguyen - Continue prototype wiring. (3)

Fearghus Tyler - Review schematic design. (3)

Yaqoub Rabiah - Continue PLC programming. (3)

Fox Kang - Continue safety-system research. (3)

Blocked

- None.
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WEEKLY REPORT 2026-02-13

Last week . (*hours in parentheses*)

Team Review - The team met with the advisor to discuss project coordination and integration with the mechanical design.

Bao Nguyen - Attended advisor meeting. (1) - Worked on electrical layout planning. (2)

Fearghus Tyler - Continued the AutoCAD schematic design — refining component blocks and wiring out the control sheet. (3) - Attended advisor meeting. (1)

Yaqoub Rabiah - Worked on PLC logic improvements. (3) - Attended advisor meeting. (1)

Fox Kang - Updated safety documentation. (3) - Attended advisor meeting. (1)

Next week

Team Plan - Continue electrical design and documentation work.

Bao Nguyen - Continue planning the panel/electrical layout — place the VFD, contactor, 24 V PSU, and branch breakers, and rough out control-wire routing for the cabinet.

Fearghus Tyler - Continue the wiring schematics; finalize the power-distribution and control sheets once the updated enclosure CAD arrives from the ME team.

Yaqoub Rabiah - Continue refining the PLC ladder logic — deadman permissive, start/stop, and I/O mapping against the points list.

Fox Kang - Continue the safety-system documentation and evaluate capacitive-sensor options for the operator-presence interlock.

Blocked

Bao Nguyen - Panel/electrical layout depends on the ME enclosure dimensions, still pending.

Fearghus Tyler - Waiting for updated CAD from ME team.

Yaqoub Rabiah - PLC I/O scaling depends on the final analog-module / VFD decision, still pending.

Fox Kang - Still selecting the capacitive-sensor technology before ordering parts.

WEEKLY REPORT 2026-02-27

Last week . (*hours in parentheses*)

Team Review - The team focused on preparing documentation and presentation materials.

Bao Nguyen - Updated electrical design diagrams. (3)

Fearghus Tyler - Reviewed component selections and revised the BOM — switched the production-motor candidate to a PE145TC-2-4 (2 HP, 145TC, 1800 RPM, 3-phase TEFC) and added the NBE7 circuit breaker. (3)

Yaqoub Rabiah - Reviewed PLC logic. (3)

Fox Kang - Updated safety documentation. (3)

Next week

Team Plan - Continue preparing final presentation materials.

Bao Nguyen - Finalize the electrical design diagrams (system block diagram and control-circuit layout) for the presentation.

Fearghus Tyler - Finalize the component-selection list / BOM (VFD, PLC, HMI, contactor, breakers, PSU) for the presentation.

Yaqoub Rabiah - Finalize the PLC control-logic write-up and clean up the ladder comments for the presentation.

Fox Kang - Finalize the safety documentation — E-stop chain, interlocks, and the accident-scenario flow chart — for the presentation.

Blocked

- None.
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WEEKLY REPORT 2026-03-06

Last week . (*hours in parentheses*)

Team Review - The team finally got the parts they requested.

Bao Nguyen - Fit parts to our prototype and wire it. (3)

Fearghus Tyler - Fit parts to our prototype and wire it. (3)

Yaqoub Rabiah - Started figuring out C-more to program the HMI. (3)

Fox Kang - Cargo delay; exploring backup plans. (2)

Next week

Team Plan - Try figuring out the VFD phases to have a working motor.

Bao Nguyen - Continue wiring the prototype and bring up the VFD + motor combo — verify phase wiring and get the motor spinning under VFD control.

Fearghus Tyler - Continue fitting and landing the newly arrived parts on the protoboard and tidy up the control wiring.

Yaqoub Rabiah - Continue learning C-more and building out the HMI screens (Ready/Running/Fault, start/stop, status display).

Fox Kang - Pursue a capacitive-touch backup approach while waiting on the chip — breadboard a simple touch sensor and sketch the X7 stop interface.

Blocked

Bao Nguyen - Waiting on E-stop switch.

Fearghus Tyler - Schematic finalization waiting on the ME team's enclosure dimensions.

Yaqoub Rabiah - Still getting C-more set up to program the HMI.

Fox Kang - Cargo delay on capacitive-touch parts.

WEEKLY REPORT 2026-03-13

Last week . (*hours in parentheses*)

Team Review - Started wiring up our prototype and getting the VFD to work.

Bao Nguyen - Work on the VFD and motor combo. (3)

Fearghus Tyler - Work on the VFD and motor combo. (3)

Yaqoub Rabiah - Continued developing the HMI GUI. (3)

Fox Kang - Get the capacitive-touch chip and prepare for the demo. (3)

Next week

Team Plan - Finalize the documentation before spring break.

Bao Nguyen - Continue VFD/motor bring-up — tune the basic run parameters — and document the wiring/parameter setup before spring break.

Fearghus Tyler - Continue the VFD/motor wiring and update the schematics to match the as-built prototype.

Yaqoub Rabiah - Continue developing the HMI GUI — refine the screen layout and link the tags to the PLC.

Fox Kang - Prepare the capacitive-touch chip for the demo — wire it up and verify a basic touch trip.

Blocked

- None.

WEEKLY REPORT 2026-03-16

Last week . (*hours in parentheses*)

Team Review - The team finalized project documentation and presentation materials.

Bao Nguyen - Updated diagrams and documentation. (3)

Fearghus Tyler - Reviewed schematic documentation. (2)

Yaqoub Rabiah - Reviewed PLC control-logic documentation. (2)

Fox Kang - Reviewed safety documentation. (2)

Next week

Team Plan - Use connectors to make it easy for the ME team to install onto the shredder. - With the extra time we have, we can implement other kinds of safety sensors.

Bao Nguyen - Add quick-connect connectors at the panel boundary so the ME team can plug in the electrical system during shredder install.

Fearghus Tyler - Update the schematics and wiring diagrams to show the connectorized field/cabinet interface.

Yaqoub Rabiah - Continue the PLC/HMI documentation and keep the points list in sync with the logic.

Fox Kang - Use the extra time to explore additional safety sensors (e.g., lid interlock, light curtain) beyond the capacitive touch.

Blocked

- None.

Spring break between 2026-03-16 and 2026-04-03.

TERM
Spring Term
April – June 2026

Universal Plastic Shredder V2.0 – Weekly Progress Reports

WEEKLY REPORT

Week 1 | 2026-04-03

Last week . *(hours in parentheses)*

Team Review - Refocused on getting back on track after spring break (3 hours)

Bao Nguyen (4) - Redoing the points list in the sponsor's preferred style — `PLC/Point_List.xlsx` is the source of truth for every X/Y and analog point. (4) - Installed and wired the E-stop button (Eaton **E22B1** NC mushroom) in series with the **SD-N35** contactor coil A1 (A2 -> PSU V); PLC **X4** taps the A1 node to monitor the chain. (1)

Fearghus Tyler - Continued the AutoCAD wiring schematics/model to match the points list — keeping the component blocks in sync with the I/O map. (2) - Continued support in wiring and design aspects of the protoboard. (1) - Revised 3D model to accommodate sponsor feedback. (1)

Yaqoub Rabiah - Figured out a rework of the HMI GUI. Concept wasn't liked by the sponsor. Made it sleeker in concept. (3 hours)

Fox Kang - Working on capacitive-touch concept. - Researched capacitive-sensing approaches for the operator-presence safety feature using an ESP32; scoped a touch-to-stop interlock that drops the run command via PLC input **X7**. (3 hours)

Next week

Team Plan - Start integrating the PLC, VFD, and HMI together for our demo. - Start materializing the final report. - Start prototyping the capacitive-touch feature.

Bao Nguyen - Start integrating PLC with the circuit to turn on and control the VFD.

Fearghus Tyler - Set up the VFD to work using the analog and digital inputs instead of the default factory settings (0–10 V). - Coordinated with the industry sponsor controls team to confirm preferred termination conventions. - Cross-reference motor data against VFD.

Yaqoub Rabiah - Start integrating PLC with the circuit to turn on and control the VFD.

Fox Kang - Editing code comments and preparing connecting the module to the PLC.

Blocked

Bao Nguyen - PLC 24 V module not suitable for contactor and LED control — need to find a relay module instead.

Fearghus Tyler - Need to know where the connections will be before finishing the schematics.

Yaqoub Rabiah - VFD speed-reference/RPM control wiring pending the analog-module configuration.

Fox Kang - Capacitive-touch concept still pending feedback from Ichor's EE/controls team before parts selection.

WEEKLY REPORT

Week 2 | 2026-04-10

Last week . *(hours in parentheses)*

Team Review - Made progress towards the capacitive-touch feature; works on a small scale. (5 hours) - Integrated the PLC to the circuit. Contactor turns on with deadman switches, as well as the VFD. (8 hours)

Bao Nguyen - Rewired the PLC and contactor for clean integration — moved the **SD-N35** coil out of the PLC output into the hardwired E-stop chain (previously driven by D2-08TD2 Y0). (4 hours) - Identified the problematic module — the **F2-06AD-1** 4-20 mA current-input on the speed-reference path wasn't behaving; flagged it for a swap. (4 hours)

Fearghus Tyler - Tested the protoboard with the analog and digital inputs. Ready to test with HMI. (2) - Created some 2D wiring diagrams of different versions for field vs. internal cabinet wiring. (2)

Yaqoub Rabiah - Reprogrammed the logic code for the PLC, ensuring the bits corresponded to the physical circuit. (2 hours) - Debugged the deadman and start/stop functions to specs while integrating the PLC. (4 hours) - Laid the foundation for the weekly progress reports. (3 hours)

Fox Kang - Prototyped the ESP32 capacitive-touch sensor on a breadboard; confirmed it triggers reliably on a small test surface. (5 hours) - Began wiring the module output toward the PLC **X7** input for the touch-to-stop interlock.

Next week

Team Plan - Continue integrating the PLC, VFD, and HMI together for our demo. - Start materializing the final report.

Bao Nguyen - Continue integrating PLC with the circuit to turn on the VFD.

Fearghus Tyler - Work with the industry sponsor to schedule a controls-team review of our materials. - Investigated VFD behavior under current control vs. voltage control. Documented the odd current readings off the module (possibly broken?).

Yaqoub Rabiah - Continue reworking the logic and HMI design for further integration.

Fox Kang - Tune sensitivity and debounce on the cap-touch prototype; reduce false triggers before PLC integration.

Blocked

Bao Nguyen - Waiting on the relay module to properly interface the PLC output with the contactor and indicator LEDs.

Fearghus Tyler - Need to know their final motor specs.

Yaqoub Rabiah - RPM control is acting weird.

Fox Kang - Cap-touch prototype gets false triggers from electrical noise near the panel; needs filtering before PLC integration.

WEEKLY REPORT

Week 3 | 2026-04-17

Last week . *(hours in parentheses)*

Team Review - Integrating PLC, HMI, VFD and motor. (8 hours)

Bao Nguyen - Investigated why the 12–24 VDC transistor output module (**D2-08TD2**) was not behaving like a proper on/off switch. Discovered that although it functions as a switch, the lowest output voltage is still too high to reliably control devices that require a clean digital signal, such as the contactor. (4) - Located a **D2-08TR** 8-ch relay-output module (dry contact, sinking, common to PSU V) in the Power Lab and wired it in to properly interface the PLC output with external loads. Tested it using the on-module indicator lights to confirm correct switching. (2) - Began studying the analog modules — **F2-08AD-1** (4–20 mA current in, Slot 3) and **F2-08DA-2** (0–10 V out, Slot 6) — to work out their wiring, scaling, and integration method. (6)

Fearghus Tyler - Proposed switching from current control to voltage control (0–10 V) to the team and sponsor. (1) - Updated 2D schematic to reflect planned analog module swap. (2) - Revised 3D control-panel model to show new voltage-module placement and wire routing.

Yaqoub Rabiah - Integrated a scale for RPM control from the HMI to the PLC. (2 hours) - Added forward and reverse capabilities into PLC logic and integrated it into the HMI. (5 hours) - Reworked the numerical displays on the HMI main page — a voltage / current / frequency / RPM read-out section (some fields not yet operational, pending the analog/telemetry scaling). (2 hours) - Debugged the connection between the PLC and HMI. The protocols weren't matching. (2 hours) - Found a problem with the current-control module and tried to debug. (4 hours)

Fox Kang - Continued bench prototyping of the cap-touch module; worked on suppressing false triggers from electrical noise near the panel. (4 hours)

Next week

Team Plan - Figure out the problem of the low amperage variance in the current-control module.

Bao Nguyen - Continue studying the analog module — determine the correct field wiring and the PLC scaling so the VFD speed reference and motor-current feedback read correctly.

Fearghus Tyler - Reprogrammed VFD parameters from current-control to voltage-control (0–10 V) following the analog-module swap — PD002=1 for the **VI** 0–10 V speed reference, motor-current monitor on **VO** via PD054=1. (3) - Bench-tested VFD response to 0–10 V signal from PLC; verified linear RPM control within expected range. (2) - Updated schematics and 2D wiring diagrams to reflect final analog module configuration and E-stop logic paths. (3) - Generated updated 3D enclosure model for ICHOR's final design package. (2)

Yaqoub Rabiah - Fix logic for the cc module.

Fox Kang - Prepare cap-touch module output for wiring to PLC X7 and test the touch-trip stop.

Blocked

Bao Nguyen - Currently working on understanding how the analog module is wired and configured within the PLC system. Still determining the correct wiring for the analog inputs/outputs and how the signals should be scaled and interpreted by the PLC. Further testing and documentation review are needed before integration with the rest of the system.

Fearghus Tyler - Troubleshooting current-controlled VFD motor control; running into issues with ranges.

Yaqoub Rabiah - Can't seem to program the current-control module such that the range is more than 0–0.3 mA.

Fox Kang - Can't wire the cap-touch module to X7 until the PLC integration is sorted out.

WEEKLY REPORT

Week 4 | 2026-04-24

Last week . *(hours in parentheses)*

Team Review - Swapped current control with voltage control module, thus ending current-control confusion.

Bao Nguyen - Swapped the current-input module for the **F2-08DA-2** 0–10 V analog-output module (drives VFD **VI**) and read the configuration docs. Found the V-memory bits holding the scaled RPM from the Do-more scaling instruction (with Yaqoub). (4) - Working with Fearghus on reprogramming the VFD. (2)

Fearghus Tyler - Tested and reprogrammed the VFD to match the move from the current-controlled to the voltage-controlled module — set PD002=1 and verified 0 V = 0 Hz, 10 V = max frequency on **VI**. (4) - Prepare any remaining VFD/PLC/HMI integration specs before demo.

Yaqoub Rabiah - Reprogrammed the PLC to accommodate the new module. (2 hours) - Added LED logic — **Y0** green (forward) / **Y1** yellow (reverse) — and swapped bits for easier wiring (VFD **FOR=Y6**, **REV=Y7**, **RST=Y5**, sinking to DCM). (1 hour) - Added E-stop logic (monitored on **X4**) and a warning-light output (**Y4** -> Wamco 525, 28 VDC) for safety. (1 hour)

Fox Kang - Refined cap-touch firmware/code comments and prepped the module for hand-off to PLC integration. (3 hours)

Next week

Team Plan - Do the presentation.

Bao Nguyen - Support presentation prep and start sourcing the remaining parts — relay module, enclosure, and braking resistor.

Fearghus Tyler - Figure out any last-minute parts we could integrate. - Investigate braking resistor.

Yaqoub Rabiah - Get ready for presentation.

Fox Kang - Prepare cap-touch demo material for the presentation.

Blocked

Bao Nguyen - None — presentation week, no blockers.

Fearghus Tyler - None.

Yaqoub Rabiah - None.

Fox Kang - Awaiting the final cap-touch parts/relay before the full PLC hand-off.

WEEKLY REPORT

Week 5 | 2026-05-01

Last week . *(hours in parentheses)*

Team Review - Did presentation. (2 hours) - Prototyped capacitive touch. (2 hours)

Bao Nguyen - Worked with Fox to tune and build a prototype of the CTSI capacitive-touch safety interface. (2) - Identified new parts needed — a dynamic braking resistor (ATO **APCS-300R30-AD**, 300 W / 30 ohm on VFD **P+ / PR**) and the enclosure.

Fearghus Tyler - Sized the dynamic braking resistor (~30 ohm, 300 W for VFD **P+ / PR**) — landed on the ATO **APCS-300R30-AD**. (1) - Got started on a user manual. (2) - Receiving final supplies from industry sponsors (enclosure, braking resistor, relay for integrating capacitive touch). (1)

Yaqoub Rabiah - Worked on presentation slides. (1 hour)

Fox Kang - Worked with Bao to tune the ESP32 CTSI capacitive-touch prototype (still on a bread-board); confirmed the touch-to-stop concept on the bench. Started planning the move to a soldered, enclosed module. (2 hours)

Next week

Team Plan - Work on enclosure.

Bao Nguyen - Continue identifying and ordering remaining parts (braking resistor, enclosure, relay) and help start the enclosure build.

Fearghus Tyler - Build and test.

Yaqoub Rabiah - Try integrating overcurrent signal.

Fox Kang - Continue developing the CTSI capacitive-touch module toward panel integration — finalize the sensor circuit and the X7 interface.

Blocked

Bao Nguyen - Can't progress on the enclosure build until the enclosure and remaining parts (braking resistor, relay) arrive from the sponsor.

Fearghus Tyler - We can't test a lot of the functionality we had aimed for due to no shredder prototype built yet.

Yaqoub Rabiah - Waiting for parts.

Fox Kang - Waiting on the relay for integrating capacitive touch.

WEEKLY REPORT

Week 6 | 2026-05-08

Last week . *(hours in parentheses)*

Team Review - Not much (midterm exam week).

Bao Nguyen - Nothing — exam week.

Fearghus Tyler - Midterm exams took priority; we mostly worked on documentation. - Minor schematic changes based on the PLC logic for switching.

Yaqoub Rabiah - Not much also (exam week).

Fox Kang - Exam week — minimal progress; reviewed cap-touch integration plan.

Next week

Team Plan - Work on enclosure setup.

Bao Nguyen - Resume parts sourcing and enclosure prep after exams.

Fearghus Tyler - Get the last parts from Ichor Systems.

Yaqoub Rabiah - Try integrating overcurrent signal.

Fox Kang - Resume cap-touch integration once the relay arrives.

Blocked

Bao Nguyen - Exam week — no project time; also waiting for parts.

Fearghus Tyler - Waiting for parts.

Yaqoub Rabiah - Waiting for parts.

Fox Kang - Waiting for parts.

WEEKLY REPORT

Week 7 | 2026-05-15

Last week . *(hours in parentheses)*

Team Review - Set up the control panel to go inside the enclosure. (5 hours) - Worked on the overcurrent-signal integration. (10 hours)

Bao Nguyen - Helped lay out and wire the control panel into the enclosure — landed 12 AWG THHN power runs (rated 20/25/30 A at 60/75/90 degC per NEC 310.16; motor FLA ~ 7.6 A) and ferruled 16–18 AWG control wiring; supported the overcurrent signal into **X5**. (12 hours)

Fearghus Tyler - We got our enclosure, so I worked on the CAD designs for the panel design and additional panel DWGs. (4) - Did a lot of troubleshooting and design changes for a clean enclosure. (3) - Helped Fox periodically with the cap-touch module. (1)

Yaqoub Rabiah - Wrote logic that takes the overcurrent signal and runs an unjam algorithm to clear the shredder — VFD fault relay **FA/FC** programmed PD052=02 -> +24 V into PLC **X5**; the routine pulses **Y5** -> **VFD RST** (~500 ms) with a 3-strike lockout. (10 hours) - Helped design the enclosure layout. (1 hour) - Drilled holes on the panel that holds the major components (VFD, contactor, circuit breakers, etc.). (1 hour) - Helped wire the components in the enclosure. (2 hours)

Fox Kang - Worked on the ESP32 CTSI module with Fearghus's help toward wiring it into panel **X7**; hit a bug where it works powered from a PC (USB) but bugs out standalone on the wall-outlet supply — a grounding / clean-power issue. Began soldering the module off the breadboard. (4 hours)

Next week

Team Plan - Finish up the enclosure setup and make sure it works — plug 'n' play.

Bao Nguyen - Continue enclosure wiring and verify panel integration — confirm relay-driven contactor switching and the overcurrent signal into the PLC.

Fearghus Tyler - Work on the final documentation / user manual / designs.

Yaqoub Rabiah - Finish enclosure setup. - Make sure logic works. - Set up cap touch.

Fox Kang - Integrate and test the cap-touch module on X7 in the enclosure.

Blocked

Bao Nguyen - Can't validate the overcurrent trip threshold without a real shredding load — no shredder prototype to test on yet.

Fearghus Tyler - Still no prototype shredder to test on.

Yaqoub Rabiah - Overcurrent-clear algorithm can't be tuned against real jam currents until a prototype exists; limited to bench simulation.

Fox Kang - CTSI module works on PC USB but bugs out on the standalone wall supply (grounding/noise); needs a stable supply before it can trip **X7** reliably.

WEEKLY REPORT

Week 8 | 2026-05-22

Last week . (*hours in parentheses*)

Team Review - Continued enclosure wiring and panel troubleshooting; coordinated parts hand-off with the mechanical team. - Roughly 20 team-hours this week.

Bao Nguyen - Continued wiring/troubleshooting the control panel and verifying PLC I/O against the points list. (14 hours)

Fearghus Tyler - Worked on wiring the panel / troubleshooting. (5) - Helped the mech team get their parts (at long last). (3)

Yaqoub Rabiah - Continued debugging PLC logic against the wired panel; verified overcurrent-clear routine behavior. (6 hours)

Fox Kang - Built up the CTSI module — received the 24 V->3 V converter (to run the ESP32 off the panel 24 V), an optocoupler to isolate the output into **X7**, and an enclosure box; soldering it together off the breadboard. Accidentally fed 24 V into the ESP32 while wiring and fried it; sourced and installed a replacement. (4 hours)

Next week

Team Plan - Finish the panel/enclosure wiring and shift focus to final documentation — final report, user manual, and schematics.

Bao Nguyen - Finalize the as-built wiring documentation and points list, and wrap up any remaining panel work.

Fearghus Tyler - Help assemble the first edition of the shredder.

Yaqoub Rabiah - Document the PLC control logic and HMI, and verify the logic on the bench setup.

Fox Kang - Finalize and document the capacitive-touch module and bench-test the touch trip.

Blocked

Bao Nguyen - Integrated/load testing still blocked — no completed shredder; work limited to bench checks and documentation.

Fearghus Tyler - Waiting on the ME team to finish the shredder frame before final install and testing.

Yaqoub Rabiah - Can't validate the control logic under real motor load without the mechanical prototype.

Fox Kang - Replacement ESP32 in and powered via the 24 V->3 V converter; still finishing the soldered/enclosed build and verifying the optocoupler trips **X7** cleanly.

WEEKLY REPORT

Week 9 | 2026-05-30

Last week . (*hours in parentheses*)

Team Review - Focused on documentation — drafting the final report, the user manual, and exporting the final schematics/diagrams for the report appendix.

Bao Nguyen - Wrote the EE sections of the final report (wiring, sizing, safety) and finalized the as-built wiring documentation and points list. (16 hours)

Fearghus Tyler - Continued the final documentation and user manual; exported the final schematics and the 3D enclosure model for the design package. (8 hours)

Yaqoub Rabiah - Documented the PLC control logic and HMI screens; wrote up the control-logic section of the report. (8 hours)

Fox Kang - Finished the soldered, enclosed CTSI module (ESP32 + 24 V->3 V converter + opto-coupler to **X7**) and documented its design for the report. (6 hours)

Next week

Team Plan - Finish and submit the final report, complete the user manual, and prepare the end-of-term demo/poster.

Bao Nguyen - Finalize the wiring documentation (as-built diagrams, points list) and complete the EE sections of the final report.

Fearghus Tyler - Finish user manual and export final schematics/DWGs for the report appendix.

Yaqoub Rabiah - Finalize PLC/HMI documentation and write up the control-logic section.

Fox Kang - Document the cap-touch module and write up its section.

Blocked

Bao Nguyen - Final end-to-end/load testing still not possible — no completed shredder; report documents bench results only.

Fearghus Tyler - Mechanical prototype still not built, so the install-and-test parts of the user manual are based on the bench setup.

Yaqoub Rabiah - Can't capture real run/jam data for the report without a working shredder; the analog current monitor (VFD **VO** -> **F2-08AD-1** CH1) is wired but still needs scaling/calibration.

Fox Kang - Couldn't demonstrate the cap-touch trip on the full machine — only on the bench.

WEEKLY REPORT

Week 10 | 2026-06-06

Last week . (*hours in parentheses*)

Team Review - Completed and submitted the final report and user manual, and presented at the capstone demo/expo.

Bao Nguyen - Completed EE report sections (wiring, sizing, safety) and supported the final demo. (4 hours)

Fearghus Tyler - Finalized the user manual and design package; delivered final schematics and 3D enclosure model. (8 hours)

Yaqoub Rabiah - Finalized PLC/HMI control documentation and supported the demo. (8 hours)

Fox Kang - Finalized the capacitive-touch module documentation and demonstrated the touch-to-stop safety feature. (6 hours)

Next week

Team Plan - Term complete — project handed off to the sponsor.

Bao Nguyen - N/A — end of term.

Fearghus Tyler - N/A — end of term.

Yaqoub Rabiah - N/A — end of term.

Fox Kang - N/A — end of term.

Blocked

Bao Nguyen - Completely blocked by the ME team — they have no built shredder for us to test against or mount the panel on; the final report documents bench results only.

Fearghus Tyler - None — end of term.

Yaqoub Rabiah - Telemetry was never fully implemented: the HMI has a voltage / current / frequency / RPM read-out section, but some fields are non-operational. The analog motor-current monitor (VFD **VO** -> **F2-08AD-1** CH1) is wired but only partially working — still needs scaling/calibration. Ran out of time; documented as future work.

Fox Kang - None — end of term.